

The role of endoscopy and photodynamic therapy in managing ulcerative colitis

Joan Gabrielle Basa (20881535) | PHYS 396 - Biophysics in Imaging | Dr. Brenda Yasie Lee | 03 April 2025

ABSTRACT

Ulcerative colitis (UC) is a chronic and idiopathic inflammatory bowel disease (IBD) characterized by continuous inflammation of the colon and rectum, leading to symptoms like abdominal pain, bloody diarrhea, and weight loss. While its exact cause remains unclear, it is thought to involve immune dysfunction, genetic susceptibility, and environmental factors. Early diagnosis and effective treatment are crucial to managing disease progression and reducing complications, including colon cancer. This project explores the role of endoscopy, the gold standard for diagnosing and monitoring UC, and investigates the therapeutic potential of photodynamic therapy (PDT). Advanced endoscopic techniques, such as high-definition white light endoscopy (HD-WLE) and autofluorescence imaging (AFI), enhance the detection of early mucosal changes and abnormal cell growth. PDT, a non-invasive treatment, uses photosensitizing agents and light to generate reactive oxygen species that selectively destroy inflamed tissue while sparing healthy cells. Combining advanced endoscopy with PDT aims to improve diagnostic accuracy, therapeutic outcomes, and patient quality of life, offering a promising minimally invasive option for UC management. Further research is needed to optimize PDT protocols and evaluate long-term efficacy and safety.

INTRODUCTION

Overview

Inflammatory bowel disease (IBD) encompasses a group of chronic inflammatory conditions, most notably ulcerative colitis (UC) and Crohn's disease (CD). While these two conditions share many common symptoms, such as diarrhea and abdominal pain, they show distinct characteristics: UC is characterized by mucosal inflammation in the colon and CD is characterized by transmural inflammation anywhere in the digestive tract.¹ Mucosal inflammation will preserve the five layers of the intestinal wall, while transmural inflammation results in interruptions and loss in the five layers of the intestinal wall.¹ Furthermore, UC patients will often experience bloody stools while CD patients do not.¹

Focusing on ulcerative colitis, the purpose of this article is to systematically evaluate the role of endoscopic imaging as a diagnostic, monitoring, and imaging modality—specifically highlighting advanced techniques such as high-definition white light endoscopy (HD-WLE) and autofluorescence imaging (AFI)—and explore the therapeutic potential of photodynamic therapy (PDT).

Ulcerative colitis

Ulcerative colitis is the most common type of IBD affecting the colon or large intestine.² Its course is often presented by periods of flare-ups, where new damage often occurs and patients actively experience the mentioned symptoms, and periods of remission, where the tissue starts to heal.^{2,3}

Epidemiology

UC occurrence is rising globally, with projections indicating that approximately 470,000 Canadians will be living with the condition by 2035.⁴ It also occurs across all ages, with Scandinavia and Canada reporting the highest incidence of paediatric IBD (in children less 16 years of age), while very-early-onset IBD (0–5 years old) has risen by 7.2% annually, potentially reflecting distinct pathophysiology; globally, UC incidence is consistent across sexes and ethnicities, pointing to environmental factors as key drivers of epidemiological trends.⁴

Moreover, the disease manifests in varying extents: around 30% of patients present with proctitis, where inflammation is confined to the rectum; 40% have distal colitis, which extends up

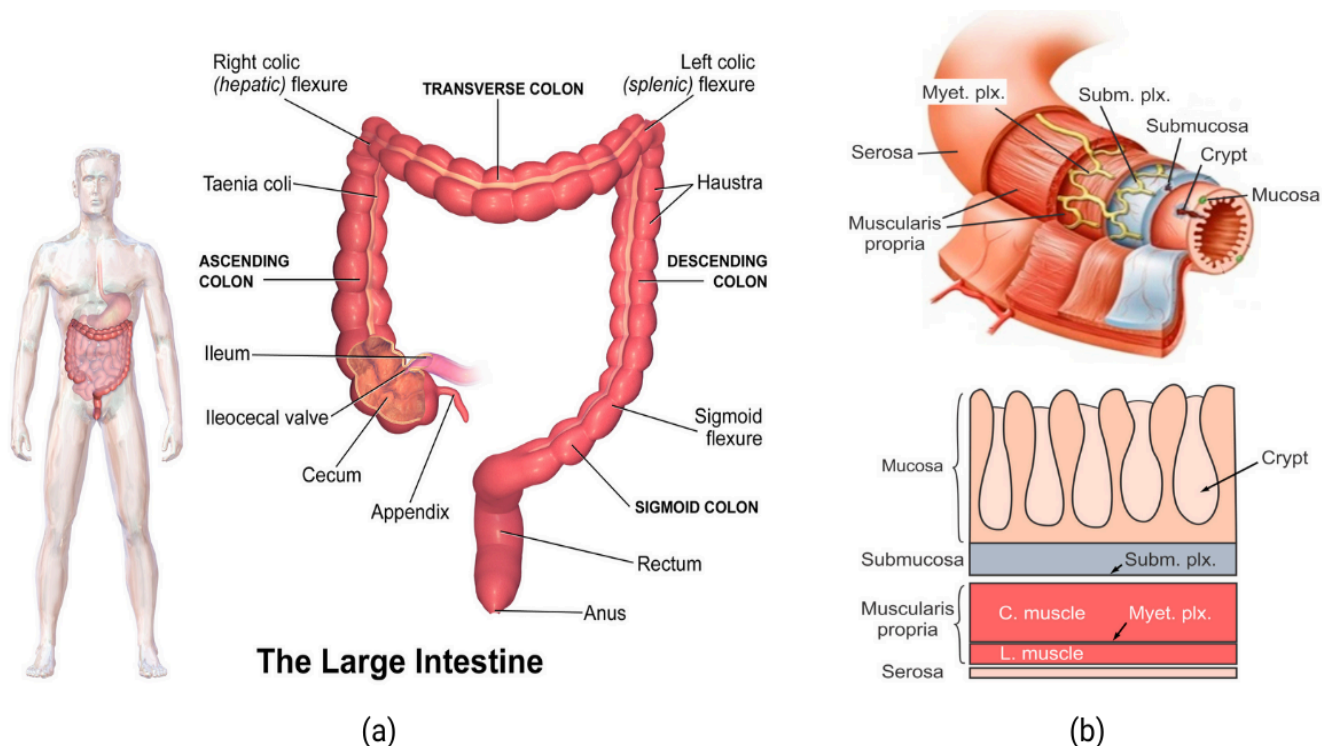


Figure 1 UC is characterized by chronic inflammation and the formation of ulcers along the inner surface (lumen) of the large intestine, including the colon and the rectum. (a) Inflammation affects the inner lining of the colon, spreading upward from the rectum and leaving no healthy tissue along its path; (b) This inflammatory process is typically confined to the mucosa and superficial submucosa.^{3,5,6}

to but not beyond the splenic flexure; and in the remaining 30%, it involves the entire colon (or what is known as pan-colitis).⁵

Etiology and pathophysiology

In general, IBD has traditionally been considered a disease entity of uncertain origin.⁷ While the precise mechanism of UC mucosal destruction remains unclear, it is believed to be multifactorial, resulting from the interplay of genetic and environmental factors, alterations in the gut microbiota, and dysregulation of the intestinal immune system.^{2,3,7,8}

Genetic and environmental risk factors

UC has a strong genetic component, with the first-degree relatives of patients having a four-fold higher risk and over 200 identified genes linked to immune responses and intestinal barrier function.⁹ Environmental factors, such as gastrointestinal infections, air pollution, and a Western diet, also contribute to its rising global incidence.^{4,9,10}

Gut microbiome

UC involves gut dysbiosis, featuring reduced microbial diversity and a huge shift in bacterial

populations.^{4,11} Sulfide-producing bacteria, linked to inflammation, are increased, whereas protective short-chain fatty acids (SCFAs), which are critical for anti-inflammatory and barrier functions, are decreased.² This gut microbial imbalance may drive UC progression by amplifying mucosal inflammation.

Immune dysregulation

Certain types of immune cells, like cytotoxic T cells, are found in the lining of the colon in people with UC.⁸ The idea is that these T cells, which are normally meant to kill infected cells, start destroying the healthy cells that line the walls of the large intestine, releasing inflammatory cytokines (e.g., TNF- α , IL-6, IL-9, IL-13, IL-23, and IL-36).^{2,4,8,10}

Current paradigms in UC evaluation and management

UC evaluation and management is a multi-faceted approach that combining clinical assessment with several key diagnostic and therapeutic modalities.^{4,9,10} The primary goal of treatment has shifted from symptomatic relief to endoscopically confirmed healing of the mucosa, with the aim of improving

long-term outcomes and reducing the risk of complications like colorectal cancer.^{4 9 10}

Colonoscopy

Colonoscopy with terminal ileum intubation (TII) (i.e., advancing the endoscope into the distal small intestine) remains as the cornerstone for UC diagnosis.^{4 7 8 9 10}

Characteristic UC endoscopic findings include the following:

- **Continuous inflammation** from the rectum, with erythema, loss of vascular pattern, friability, and ulcerations.^{7 9 10}
- **Sharp demarcation** between affected and normal mucosa.⁴

Colonoscopy evaluates the severity of the inflammation and screens for colitis-associated neoplasia (CAN).¹²

Biopsy

In establishing UC diagnosis, multiple biopsies (which includes a normal-looking mucosa) reveal characteristic histological features like crypt distortion, basal plasmacytosis, and neutrophilic infiltration while excluding infections or other forms of colitis.^{8 9 10} For dysplasia surveillance, targeted biopsies of visible lesions (especially with chromoendoscopy) are prioritized, though random biopsies may supplement high-risk cases.^{8 9 10 13 14} Histologic evaluation remains definitive for assessment of inflammation severity, detection of dysplasia, and diagnosis of cancer.¹⁰

Radiology

While radiologic imaging is not the primary diagnostic tool for UC, it is useful in specific cases. CT scans help assess complications like toxic megacolon or perforation, while MRI aids in differentiating UC from Crohn's disease by detecting narrowed or blocked areas in the intestines.^{7 9 10} Ultrasound is also increasingly used for follow-up after acute colitis episodes to assess bowel wall thickness.⁹

Surgery

Colectomy, or colon resection, is indicated for medically refractory UC, neoplasia, or other severe complications (e.g., toxic megacolon, perforation, hemorrhage).^{7 9 10} While colectomy is curative for colonic disease, some pouch-related complications (e.g., pouchitis) may persist post-operatively.^{7 9}

Medications

Treatment for UC involves a step-up approach based on disease severity. Mild-to-moderate UC typically begins with 5-aminosalicylates (5-ASA) for inducing and maintaining remission, while corticosteroids are reserved for short-term remission induction in moderate-to-severe disease but not recommended for long-term maintenance due to side effects.^{7 9 10} Immunomodulators, such as azathioprine and 6-mercaptopurine, can help maintain remission in steroid-dependent cases.^{7 9} For refractory or severe disease, biologic therapies (i.e., anti-TNF, anti-integrin, and anti-IL-12/23), JAK/calcineurin inhibitors (e.g., cyclosporine, tacrolimus, and tofacitinib) are used.^{7 9 10}

The treatment goal is steroid-free endoscopic and histological remission to improve long-term outcomes and reduce colectomy risk.^{4 7 10}

METHODS

Endoscopy

An endoscope is a long, flexible tube equipped with a light and camera that can be inserted either through the mouth (for *upper endoscopy*, visualizing the upper digestive tract) or the rectum (for *colonoscopy*, examining the lower digestive tract).^{1 2 8 16} As shown in Figure 2, the distal tip includes:

- An **air/water nozzle** for cavity irrigation¹⁵
- An **objective lens** (part of the optical system for visualization)¹⁵
- **Light guides** (transmitting light via total internal reflection)¹⁵
- A **biopsy channel** for tissue sampling¹⁵

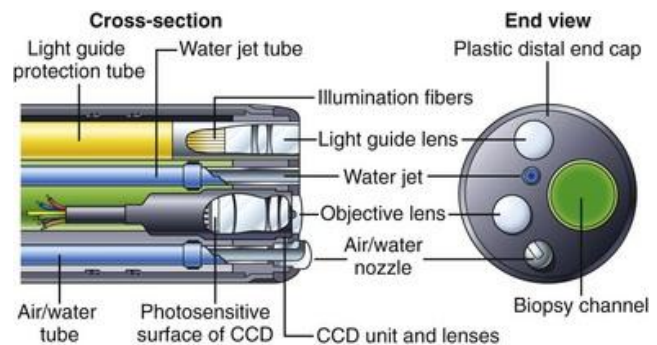


Figure 2 Distal tip assembly of a colonoscope.¹⁵

Image detection and processing begins with the built-in light source illuminating the internal tissue.¹⁵ The objective lens at the distal tip focuses a detailed image of the illuminated area onto the

charge-coupled device (CCD) sensor, which consists of a grid of light-sensitive pixels.¹⁵ Each pixel detects the brightness of the incoming light and converts it into an electrical charge through the photoelectric effect.¹⁵ These charges are stored and sequentially transferred through the CCD, where they are processed and converted into a digital image.¹⁵ The final image is then displayed on a monitor in real-time, allowing physicians to examine the internal structures with high clarity and precision for diagnostic and treatment purposes.^{8 15}

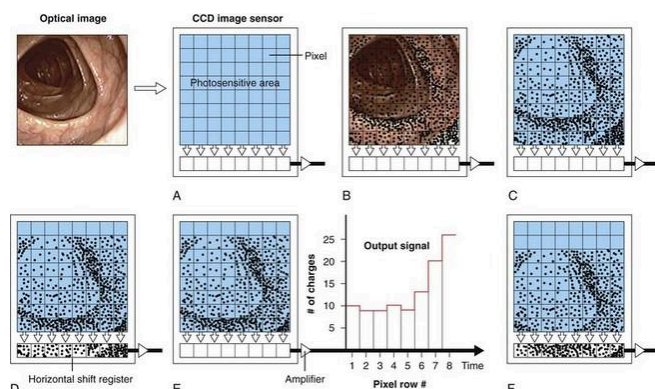


Figure 3 Image acquisition and readout using a line-transfer CCD. The process involves projecting a mucosal image onto a CCD's photosensitive surface (A), generating an electrical signal replica via the photoelectric effect (B) that persists when shielded (C), then sequentially shifting rows downward (D-F) to transfer data through a horizontal register into an output signal. Actual endoscopic CCDs have hundreds of thousands of pixels, but this illustration uses a simplified 8×8 (64-pixel) array.¹⁵

High-definition white light endoscopy

HD-WLE represents a significant advancement in UC colonoscopy as it employs high-resolution CCD sensors with over one million pixels to visualize mucosal detail.^{8 14 17} Since its clinical introduction in 1993, HD-WLE has replaced standard-definition endoscopy as the 'gold standard' for UC assessment, offering superior image resolution and magnification capabilities enabling precise evaluation of mucosal vascular patterns, subtle inflammation, and early neoplastic changes.^{8 12 14 16 17 18}

Application in UC

HD-WLE enables direct visualization of the colonic mucosa, facilitating the identification of characteristic macroscopic features of UC (e.g., obliterated vascular patterns, mucosal friability,

erythema, erosions, and ulcerations).^{4 8 10 12} These features vary in appearance depending on the severity of inflammation, allowing clinicians to classify disease activity using standardized scoring systems like the Mayo score or the Ulcerative Colitis Endoscopic Index of Severity (UCEIS).^{4 8 9 12} Through these scoring systems, clinicians can distinguish between mild, moderate, and severe colitis which is critical for adjusting treatment strategies [Figure 4].

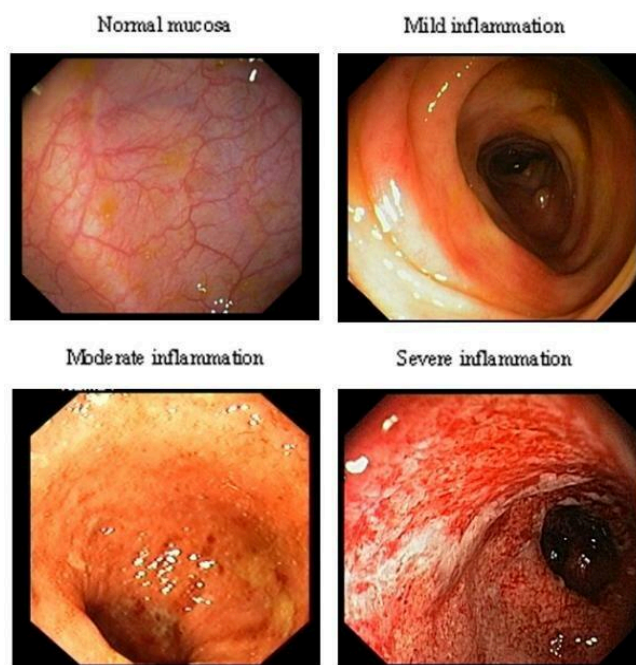


Figure 4 UC as seen from HD-WLE. Normal colon mucosa appears smooth and shiny with clearly defined vascular patterns and uniform light reflection. In mild-to-moderate UC, the mucosa demonstrates erythema, granularity, obliterated vascular patterns, and scattered erosions. Severe UC exhibits intense erythema with spontaneous bleeding, mucosal friability, ulcerations, and complete loss of vascular patterns, reflecting widespread mucosal damage.^{8 14 19}

Insights

HD-WLE offers several advantages over conventional endoscopy. Its superior image quality, with resolutions exceeding one million pixels per image, allows for detailed visualization of mucosal texture, tone, and contrast.^{8 14} This enhanced clarity not only improves diagnostic accuracy but also increases the detection rate of adenoma, a benign (non-cancerous) tumour originating from epithelial tissues, and other lesions.^{8 14 17 18} However, the procedure's efficacy relies heavily on adequate bowel preparation, as residual mucus, fluid, or plaques can obscure mucosal details.⁸

An invasive procedure, HD-WLE requires the insertion of a flexible endoscope through the anus into the colon.¹⁵ While generally well-tolerated, it may be challenging in severe cases of UC due to risk of bowel perforation.⁴ In such cases, sigmoidoscopy—a less extensive alternative limited to visualizing the rectum and sigmoid colon—may suffice for initial assessment.⁷ The complexity of HD-WLE instrumentation and methodology is manageable in most endoscopy units, though it demands skilled endoscopists to maximize its potential.¹³

From a cost and time perspective, HD-WLE represents the baseline standard for endoscopic evaluation, with associated costs and procedural durations comparable to those of conventional colonoscopy.^{8 14} Colonoscopy with HD-WLE scan duration can likely range from approximately 10 to 35 minutes, with factors like the extent of the exam and the need for additional interventions influencing the total time.^{13 14 18} While more advanced imaging techniques may offer additional insights, they often come with higher costs and extended procedural times, making HD-WLE a practical and efficient choice for routine use.¹⁴

Autofluorescence imaging endoscopy

AFI is an advanced endoscopic technique that enhances UC assessment by detecting tissue autofluorescence.^{8 12 13} Using blue light (390 to 475 nm) to excite endogenous fluorophores (e.g., collagen type-I), AFI captures their emitted long-wavelength light via CCDs, generating real-time pseudocolour images.¹² Normal mucosa appears green due to strong submucosal collagen fluorescence, while inflamed or neoplastic areas show distinct patterns [Figure 5].^{8 12 13 17} As an addition to HD-WLE, AFI improves detection of mucosal and submucosal changes, aiding in UC diagnosis and management through revealing inflammation and neoplasia with enhanced contrast.⁸

Application in UC

In UC imaging, AFI's clinical utility is twofold. First, it quantifies inflammation severity: the green autofluorescence intensity has an inverse correlation with mucosal inflammation severity, since dense inflammatory infiltrates physically block submucosal collagen's fluorescent emission [Figure 5, Ia-b vs. IIa-b].^{8 12} This enables detection of subtle inflammation missed by HD-WLE, with

RGB colour analysis providing objective severity grades.¹² Second, AFI improves neoplasia surveillance; neoplastic tissues appear deep purple or magenta due to lost collagen autofluorescence, significantly reducing lesion miss rates versus white light imaging (WLI) [see Figure 5, IIIa-b].^{13 17} By highlighting these mucosal and submucosal changes, AFI guides targeted biopsies through focusing on the most suspicious regions and minimizes the need for highly extensive random biopsies, which can be time-consuming and may still miss lesions.^{13 14}

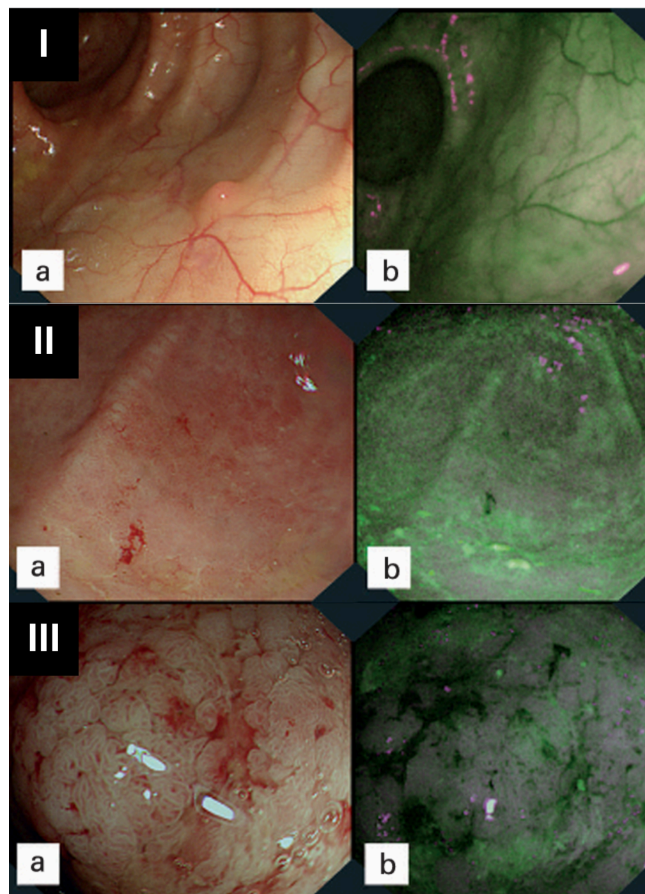


Figure 5 Endoscopic photos from white light (a) vs. autofluorescence (b) imaging. (I) Normal mucosa; (II) Inflamed mucosa with obliterated vascular patterns; (III) Mucosa with low-grade neoplasia appearing as deep purple on AFI.¹³

Insights

AFI is performed during standard colonoscopy, and its integration into modern high-definition endoscopy systems ensures high-quality imaging with pixel arrays of 640×480 in 24-bit colour.¹² The technology is user-friendly, often described as a "push-button" system, and can be seamlessly incorporated into routine endoscopic workflows.⁸ The analysis of AFI images, particularly the red,

green, and blue (RGB) colour components, can be performed using a software, further enhancing its feasibility in clinical practice.¹²

From a cost and time perspective, AFI does not impose significant additional burdens.¹³ As mentioned, its ability to improve lesion detection and guide targeted biopsies may reduce the need for extensive histopathological analysis, indirectly lowering associated costs.²⁰ However, the effectiveness of AFI still depends on adequate bowel preparation, like HD-WLE, as residual debris affects image details.¹³

Photodynamic therapy

PDT is a targeted form of treatment leveraging light-activated photosensitizers (PS) to generate localized therapeutic effects.^{21 22 23 24 25} Once administered systemically, PS will accumulate selectively in tissues and, when activated by specific wavelengths of light (usually red or blue, through lasers/LEDs), undergo photoexcitation.^{21 24} This triggers two key reactions:

- **Type I:** Produces hydroxyl ($\text{OH}^\bullet$) and superoxide ($\text{O}_2^{\bullet-}$) radicals²¹
- **Type II:** Generates singlet oxygen ($^1\text{O}_2$).²¹

These ROS induce cellular apoptosis/necrosis, offering precise control for inflammatory and neoplastic conditions.^{21 22 26 27}

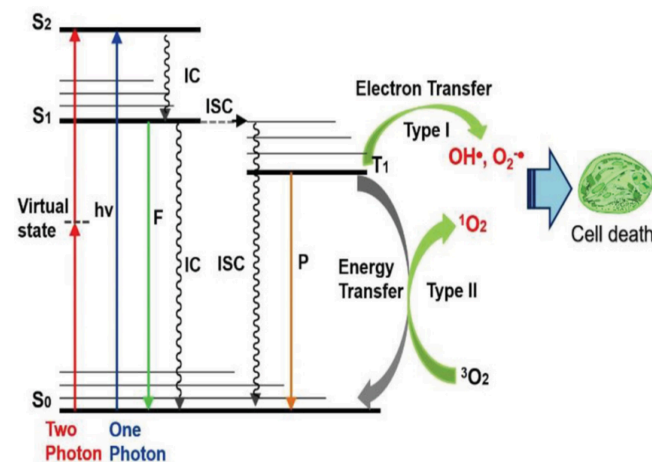


Figure 6 Production of ROS in Type I and II reactions.²¹

Application in UC

PDT is emerging as a promising targeted treatment for UC, leveraging light-activated PS to simultaneously address multiple pathogenic mechanisms.^{21 22 24} The procedure's efficacy is enhanced by the colon's tubular anatomy, which facilitates precise endoscopic light delivery via

fiber optics, enabling optimal photosensitizer activation within inflamed mucosal areas while minimizing systemic effects.²²

PDT exerts potent anti-inflammatory effects through multimodal immunomodulation. It has been shown to modulate immune responses by reducing pro-inflammatory cytokines (e.g., IL-6), while increasing anti-inflammatory cytokines (e.g., IL-10).^{11 22 28 29} This immunomodulatory activity extends to key signaling pathways, including inhibition of NF- κ B, a transcription factor that is a central regulator of inflammation.¹¹ These immunomodulatory effects have been demonstrated in animal models, where PDT significantly reduced colitis severity and promoted mucosal healing.^{11 29}

Another promising application of PDT is its ability to modulate gut microbiota, which is increasingly recognized as a critical factor in UC pathogenesis. Studies suggest that PDT can restore microbial balance by reducing harmful bacteria and promoting a healthier intestinal flora.^{11 22 28 29} This effect may be direct, through the antimicrobial properties of ROS, or indirect, resulting from the reduction of inflammation and improvement in mucosal integrity.^{22 28}

Insights

PDT is a minimally invasive procedure.^{11 21 22 27} The therapy can be delivered endoscopically, allowing precise application to affected areas while minimizing systemic toxicity.^{11 22} Advances in endoscopy imaging technology, like endoscopic capsules, hold promise for further refining the delivery of PDT, making it even less invasive and more accessible.²¹

Despite its potential, photodynamic therapy faces several challenges that must be addressed to optimize its use in UC. One key limitation is its reliance on oxygen, as severely inflamed or damaged tissues often exhibit hypoxia, thereby impairing ROS production and therapeutic efficacy.^{21 22 24 25} Effective delivery and uniform distribution of photosensitizers to target tissues, particularly in diffuse lesions, also remains a technical challenge due to mucosal irregularities.²² Furthermore, the lack of standardized protocols and the pressing need for specialized equipment and expertise may hinder the widespread adoption of PDT, especially in resource-limited settings.^{21 22 26}

With respect to cost and time, PDT adds a layer of complexity to conventional endoscopic procedures. While it offers therapeutic benefits, the need for photosensitizers, specialized light sources, and extended procedural times may increase overall costs.^{21 23 27} However, its potential to reduce inflammation could offset these expenses through decreasing the need for any additional treatments or hospitalizations.^{11 23}

RESULTS

HD-WLE is non-inferior to HD-DCE/VCE for detecting neoplasia in IBD surveillance

Iacucci et al.'s randomized and non-inferiority trial compared three techniques for detecting neoplastic lesions (i.e., abnormal cell growth) during surveillance colonoscopy in patients with longstanding IBD, including UC: high definition white light endoscopy (HD-WLE), high definition dye spraying chromoendoscopy (HD-DCE), and high definition virtual chromoendoscopy (HD-VCE) using iSCAN module image-enhanced colonoscopy.¹⁸ The study enrolled 270 patients with inactive disease who were randomly assigned to one of the three arms (90 per arm), and all colonoscopies were performed by a single experienced operator with targeted biopsies taken from detected lesions.¹⁸ The primary objective was to compare neoplastic lesion detection rates between HD-WLE with DCE and HD-WLE with VCE.¹⁸ The secondary objectives included comparing HD-WLE alone versus combined HD-WLE with either DCE or VCE.¹⁸

Key findings

Non-inferiority in detection rates

The study demonstrated non-inferiority across all comparisons: virtual chromoendoscopy (VCE) matched dye-spray chromoendoscopy (DCE) for detecting dysplasia, adenocarcinoma, and all lesions.¹⁸ Similarly, high-definition white light endoscopy (HD-WLE) was non-inferior to DCE for dysplasia/adenocarcinoma and all lesions, and to VCE for dysplasia, adenocarcinoma, and all lesions.¹⁸

Additionally, the detection rates for polypoid and non-polypoid lesions were comparable across

all three techniques [Figure 7].¹⁸ VCE showed the highest sensitivity (92.3%) but lower specificity (62.5%) for predicting neoplastic histology, while HD-WLE balanced sensitivity (91.3%) and specificity (78.1%).¹⁸ DCE had intermediate performance (sensitivity: 84.6%, specificity: 79.5%).¹⁸ Non-polypoid dysplasia detection was similar but limited by small sample sizes.¹⁸

Implications

Although HD-WLE alone may suffice for dysplasia detection in IBD surveillance when done by expert endoscopists, however, the single-operator design limits generalizability.¹⁸ Hence, multicenter trials are needed to validate these findings across various skill levels.¹⁸ Standardizing protocols (e.g., dye concentration and substitutions in DCE) and prioritizing two techniques, rather than three, may help streamline future research.¹⁸

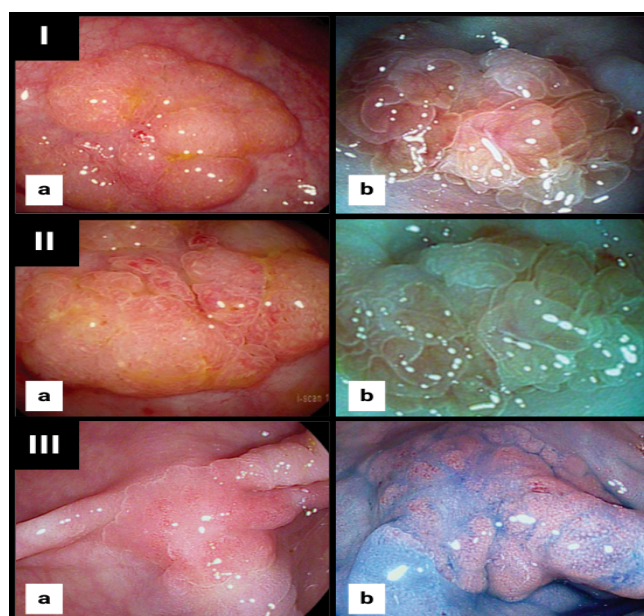


Figure 7 (I-II) Polypoid lesion with low-grade dysplasia via (a) HD-WLE showing areas of Kudo's pit pattern II–IV vs. (b) VCE showing Kudo's pit pattern III–IV; (III) Non-polypoid lesion via (a) HD-WLE exhibiting flat lesions vs. (b) HD-DCE with 0.03% methylene blue dye.¹⁸

AFI shows green colour intensity inversely correlates with UC inflammation severity

Osada et al. investigated the clinical relevance of AFI endoscopy in the assessments of UC-related inflammation, comparing it to conventional white light imaging (WLI) in 42 UC patients.¹² A total

of 572 endoscopic images (286 WLI and 286 AFI from the same sites) were analyzed, providing

valuable insights into the potential of AFI as a diagnostic tool for UC.¹²

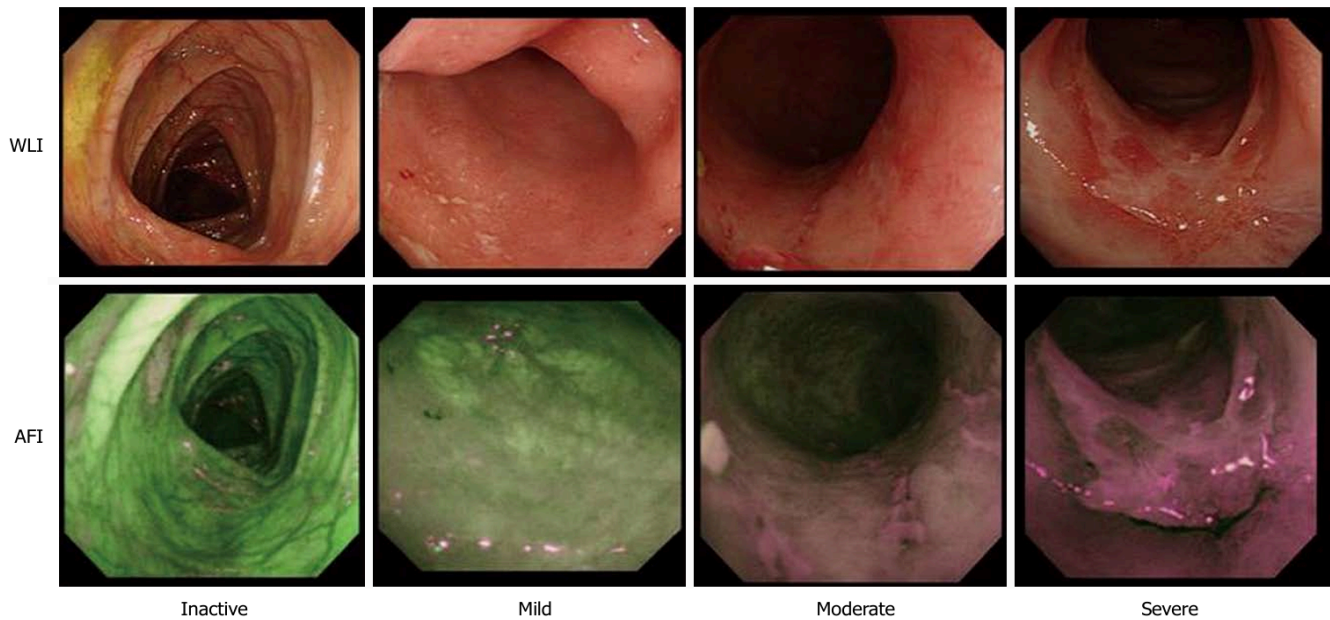


Figure 8 Paired white light imaging (top row) and autofluorescence imaging (bottom row) endoscopic images of ulcerative colitis at matching sites, with colour ranging from green to grayish and magenta, showing inactive to severe disease activity.¹²

Key findings

Quantitative assessment using the RGB colour model

The study analyzed 572 endoscopic images from 286 sites using RGB colour components.¹² The green colour component showed the strongest inverse correlation with mucosal inflammation, diminishing as inflammation severity increased.¹² Quantitative analysis revealed significant differences in green colour values across Mayo Endoscopic Subscores (MES): MES-0 (0.396 ± 0.043) vs. MES-1 (0.340 ± 0.035), MES-2 (0.324 ± 0.034), and MES-3 (0.286 ± 0.043).¹² However, as shown in Figure 7, differentiation between moderate (MES-2) and severe (MES-3) UC inflammation was limited, suggesting the green component best differentiates mild-to-moderate disease.¹²

Correlation with endoscopic and histological features

The AFI green component correlated strongly with endoscopic features, particularly vascular pattern and edema, but weakly with ulcers.¹² Histologically, it is associated best with polymorphonuclear cell infiltration and crypt

architectural distortion, but not with crypt abscesses or mucin depletion.¹² Notably, within MES-0 (endoscopically quiescent disease), AFI distinguished limited vs. extensive microscopic inflammation, highlighting its sensitivity to subclinical activity.¹²

Implications

The study found that AFI's green signal provides an objective, real-time measure of inflammation, particularly for detecting microscopic-to-moderate activity, which is a critical gap in conventional WLE.¹² While AFI cannot reliably differentiate moderate from severe inflammation (where WLE suffices), its ability to identify subclinical inflammation may improve relapse prediction and treatment decisions.¹²

LD₄-PDT alleviates UC via NF-κB pathway inhibition

Rong et al. explored the therapeutic potential and mechanisms of PDT with a novel photosensitizer, LD₄, in rat models with 3% dextran sodium sulfate (DSS)-induced UC.¹¹ UC was induced in drinking water starting on day 0, and treatments commenced on day 7.

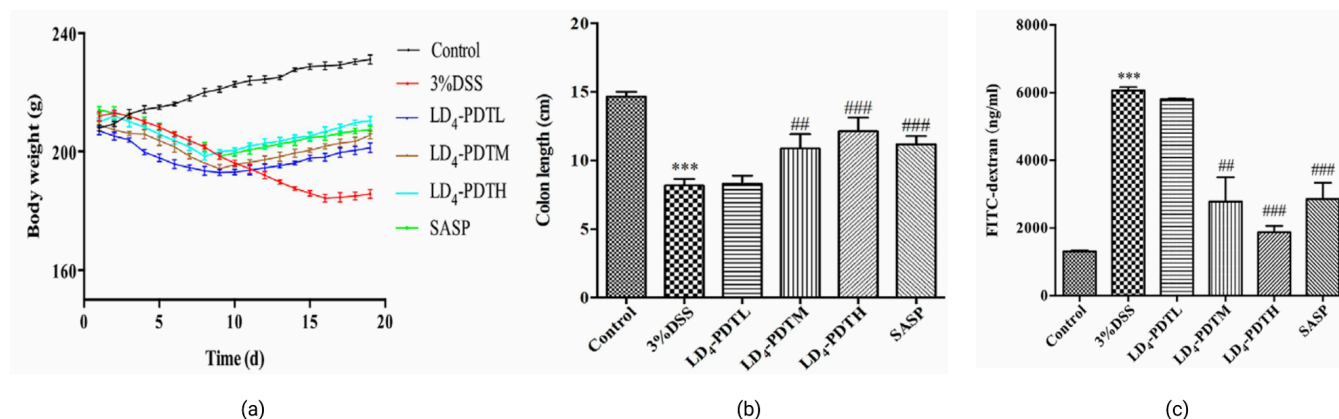


Figure 9 LD₄-PDT significantly reduced colon inflammation in DSS-induced UC rats. (a) Body weight of samples over time; (b) Colon length analysis; (c) FITC-dextran levels in rat serum.¹¹

Key findings

Therapeutic effects of LD₄-PDT

In the study, LD₄ was administered via enema every other day for a total of four treatments, followed 30 minutes later by irradiation of the colon via a semiconductor laser with a wavelength of 650 nm.¹¹ The researchers found that LD₄-photodynamic therapy (PDT) demonstrated significant protective effects in DSS-induced UC in rats.¹¹ Treatment with LD₄-PDT decreased weight loss [Figure 8a], preserved colon length [Figure 8b], and reduced intestinal permeability [FITC-dextran assay, Figure 8c].¹¹ Histologically, LD₄-PDT promoted mucosal healing, restored goblet cells, and decreased neutrophil infiltration in a dose-dependent manner, outperforming the standard sulfasalazine (SASP) treatment.¹¹ These improvements correlated with reduced expression of pro-inflammatory cytokines (MPO, IL-6, TNF- α) and oxidative stress markers (MDA), alongside increased antioxidants (GSH, SOD).¹¹

Mechanisms of action

The study found that LD₄-PDT suppressed UC progression by inhibiting the NF- κ B pathway, as shown in the reduced phosphorylation of IKK, I κ B, and NF- κ B.¹¹ Molecular docking revealed LD₄'s strong binding affinity to AOC1, a protein linked to inflammation, suggesting a potential target for PDT's anti-inflammatory effects.¹¹ Additionally, LD₄-PDT restores gut microbiota balance.¹¹ This modulation of dysbiosis, combined with its antioxidative and anti-inflammatory actions, underscores LD₄-PDT's multifaceted therapeutic potential.¹¹

Implications

LD₄-PDT emerges as a promising UC treatment by addressing these key pathological features: inflammation, oxidative stress, and gut microbiota imbalance.¹¹ Its ability to target the NF- κ B pathway and the AOC1 protein highlights a novel mechanism distinct from conventional therapies.¹¹ However, further studies are needed to validate its safety, optimize dosing, and assess efficacy in human trials.¹¹ If successful, LD₄-PDT could offer a targeted, minimally invasive alternative for refractory ulcerative colitis, particularly for patients unresponsive to current options.¹¹

CONCLUSION

Summary

UC is a disease with no permanent cure and its symptoms continue to patients' lives at an immense level, and at times, could even be life-threatening. Early intervention, preventative care, and effective therapeutics are essential for alleviating UC symptoms, improving patient quality of life, and screening for colorectal cancer. In light of managing UC, emerging techniques like HD-WLE and AFI have revolutionized inflammation surveillance, while novel treatments like PDT offer targeted approaches.

With the findings reported, HD-WLE was shown to be just as capable of detecting lesions as more advanced techniques such as DCE and VCE. However, HD-WLE is limited in its ability to detect macroscopic tissue changes—AFI can address this need, which can be crucial for treatment planning and maintaining remission. PDT with the novel photosensitizer LD₄ showed

promising dose-dependent restorative effects comparable to the conventionally-used SASP drug, with limited side effects. The advancements in these diagnostic and therapeutic modalities collectively pave the way for more precise and effective UC management, drastically improving patient outcomes.

Future directions

Multicenter trials

Greater efforts are needed to conduct multi-center endoscopy and PDT trials to establish more precise, standardized, and uniform practices for UC imaging and treatment. Current evidence for advanced endoscopic imaging and photodynamic therapy (PDT) in UC largely stems from single-center studies conducted by specialized operators. While these studies demonstrate promising results, their generalizability and applicability to real-world clinical settings remains uncertain. The lack of multicenter validation raises concerns about whether results can be replicated in routine clinical practice.

Biomarkers

Another important practical implication is leveraging biomarker-driven approaches to enhance therapy precision in IBD management.³⁰ Various molecular imaging techniques like fluorescently-labeled probes, enable targeted detection of dysplasia and colitis-associated cancer during endoscopy.^{17 30} These probes, combined with confocal laser endomicroscopy (CLE), allow real-time visualization of molecular targets (e.g., c-Met) in dysplastic tissue, improving characterization of lesions.^{17 30}

PDT leverages similar biomarker principles for selective therapy with PS accumulate in dysplastic tissue, serving as both diagnostic biomarkers and therapeutic agents for localized PDT.^{11 23} Emerging strategies explore receptor-targeted PS to enhance drug specificity.²⁴ Post-PDT, changes in inflammatory biomarkers (e.g., cytokine levels) could also aid in monitoring treatment response.^{22 26 27 28}

Fecal biomarkers (calprotectin, lactoferrin) are also increasingly used to detect and monitor intestinal inflammation.^{2 4 7} Elevated levels of these biomarkers have been found to correlate with the presence and severity of mucosal

inflammation in IBD.^{2 4} They can also be used to monitor disease activity and response to treatment, and even predict symptomatic flares.⁴

Alternative delivery systems

Alternative delivery systems for photosensitizers can further improve targeting and reduce side effects. Nanocarriers like liposomes and polymeric or inorganic nanoparticles enhance PS delivery through increased vascular permeability and receptor-mediated uptake, particularly in neoplastic tissues.²⁴ Liposomes can orient PS toward low-density lipoprotein receptors, which are overexpressed in tumours, while polymeric nanoparticles protect PS from degradation and allow ligand attachment for precise targeting.²⁴ Innovative platforms such as DNA origami, extracellular vesicles, and aggregation-induced emission nanovesicles offer programmable, biocompatible delivery with enhanced tissue penetration.^{22 24} Beyond nanocarriers, PS can be conjugated with targeting ligands like antibodies or proteins to bind tumour-specific antigens, or combined with physical methods (e.g., electroporation) to boost cellular uptake.^{21 24} Endoscopic delivery techniques include intraluminal light application via laser fibers after intravenous PS injection, or topical administration of agents like 5-aminolevulinic acid directly to mucosal lesions.^{24 26 27 31} In addition, ingestible capsules with integrated light sources promise to extend PDT to hard-to-reach areas like the small bowel, further minimizing invasiveness.²¹

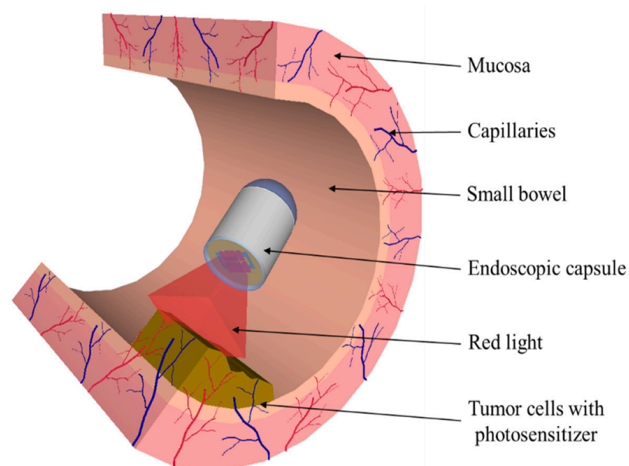


Figure 10 PDT-integrated endoscopic capsule.²¹

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